

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
2nd Semester of B. Pharm. Examination (Old Syllabus)
University Theory Examination April/ May 2016
PH109 Pharmaceutical Engineering-II

Date: 03/05/2016, Tuesday Time: 10:00 a.m to 01:00 p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

- Q 1** Attempt any **TWO** of the following
- A 1. Define filtration. Give its mechanisms. 05
 2. Give the factors affecting rate of filtration. 05
- B 1. Define centrifugation. Discuss principle and theory of centrifugation. 05
 2. Explain the construction, working and application with labeled diagram of Evaporating Pan. 05
- C 1. Discuss briefly the role of Duhring's rule and Raoult's law. 05
 2. Differentiate multiple effect & single effect evaporator. 05
- Q 2** Attempt any **TWO** of the following
- A 1. Explain Plate and Frame filter press with labeled diagram. 05
 2. Briefly describe the filter aids. 05
- B 1. How film evaporators are differentiated from other conventional evaporators on the basis of their working principle? 05
 2. Describe rotary drum filter with diagram. 05
- C 1. Write the construction and working of a perforated basket centrifuge. 05
 2. Give the applications of centrifugation in pharmacy. 05

SECTION- II

- Q 3** Attempt any **TWO** of the following.
- A Explain the principle, construction, working and advantages of fluidized bed dryer with neat diagram. 10
- B Describe the principle, construction, working, advantages, disadvantages and uses of flash distillation. 10
- C Explain the principle and working of air conditioner with neat diagram. 10
- Q 4** Attempt any **TWO** of the following.
- A Explain the construction, working, advantages and disadvantages of spray dryer. 10
- B Write the applications, theory and principle of refrigeration cycle. 10
- C What is molecular distillation? Explain the principle, construction and working of centrifugal molecular still. 10